

Randomized Algorithms (2026 Spring)

Quiz 4 – Detailed Solutions

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Problem 1

Problem. Consider the two-state Markov chain with transition matrix

$$P = \begin{pmatrix} p & 1-p \\ 1-p & p \end{pmatrix}.$$

Find a simple expression for $P_{0,0}^t$.

Solution

There are two states, which we call 0 and 1. At every step, the chain stays at its current state with probability p , and switches to the other state with probability $1 - p$.

We give two derivations of the same formula.

Method 1: Counting the parity of the number of switches

Starting from state 0, the chain is back at state 0 after t steps if and only if the number of switches among these t steps is even. Let Y be the number of switches in the first t steps. Since each step switches with probability $1 - p$, independently of the other steps,

$$Y \sim \text{Bin}(t, 1 - p).$$

Therefore,

$$P_{0,0}^t = \Pr[Y \text{ is even}] = \sum_{\substack{j=0 \\ j \text{ even}}}^t \binom{t}{j} (1-p)^j p^{t-j}.$$

We now simplify this even-indexed binomial sum carefully. For real numbers a and b , the binomial theorem gives

$$(a + b)^t = \sum_{j=0}^t \binom{t}{j} a^j b^{t-j}.$$

Also,

$$(b - a)^t = \sum_{j=0}^t \binom{t}{j} (-a)^j b^{t-j} = \sum_{j=0}^t (-1)^j \binom{t}{j} a^j b^{t-j}.$$

Adding these two identities term by term, we obtain

$$(a + b)^t + (b - a)^t = \sum_{j=0}^t (1 + (-1)^j) \binom{t}{j} a^j b^{t-j}.$$

If j is odd, then $1 + (-1)^j = 0$, so the odd terms disappear. If j is even, then $1 + (-1)^j = 2$, so the even terms are doubled. Hence

$$(a + b)^t + (b - a)^t = 2 \sum_{\substack{j=0 \\ j \text{ even}}}^t \binom{t}{j} a^j b^{t-j}.$$

Dividing by 2 gives the identity

$$\sum_{\substack{j=0 \\ j \text{ even}}}^t \binom{t}{j} a^j b^{t-j} = \frac{(a + b)^t + (b - a)^t}{2}.$$

Notice that the second term is $(b - a)^t$, not $(b + a)^t$; the minus sign is what cancels the odd terms.

Now take

$$a = 1 - p, \quad b = p.$$

Then

$$a + b = 1, \quad b - a = p - (1 - p) = 2p - 1.$$

Therefore,

$$P_{0,0}^t = \frac{1^t + (2p - 1)^t}{2} = \frac{1 + (2p - 1)^t}{2}.$$

Thus

$$\boxed{P_{0,0}^t = \frac{1 + (2p - 1)^t}{2}}.$$

This also works when $t = 0$, since $P_{0,0}^0 = 1$ and the formula gives $(1 + 1)/2 = 1$.

Method 2: Diagonalizing the transition matrix

We now derive the same formula by diagonalizing P in full detail. Let

$$q = 1 - p.$$

Then

$$P = \begin{pmatrix} p & q \\ q & p \end{pmatrix}.$$

Step 1: Find the eigenvectors and eigenvalues. Consider

$$u_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Then

$$P u_1 = \begin{pmatrix} p & q \\ q & p \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} p + q \\ q + p \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = u_1,$$

because $p + q = 1$. Hence u_1 is an eigenvector with eigenvalue 1.

Next consider

$$u_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Then

$$P u_2 = \begin{pmatrix} p & q \\ q & p \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} p - q \\ q - p \end{pmatrix} = (p - q) \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Since

$$p - q = p - (1 - p) = 2p - 1,$$

we get

$$Pu_2 = (2p - 1)u_2.$$

Thus the two eigenvalues are

$$\lambda_1 = 1, \quad \lambda_2 = 2p - 1.$$

Step 2: Form the diagonalization. Put the two eigenvectors as the columns of a matrix:

$$S = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Then the corresponding diagonal matrix of eigenvalues is

$$D = \begin{pmatrix} 1 & 0 \\ 0 & 2p - 1 \end{pmatrix}.$$

The matrix S has determinant

$$\det(S) = (1)(-1) - (1)(1) = -2,$$

and hence

$$S^{-1} = \frac{1}{-2} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Since the columns of S are eigenvectors, we have

$$PS = SD.$$

Multiplying on the right by S^{-1} gives

$$P = SDS^{-1}.$$

Step 3: Raise the matrix to the t th power. Using $P = SDS^{-1}$ and $S^{-1}S = I$, we get

$$P^t = (SDS^{-1})^t = SD^tS^{-1}.$$

Here

$$D^t = \begin{pmatrix} 1^t & 0 \\ 0 & (2p - 1)^t \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & (2p - 1)^t \end{pmatrix}.$$

Let

$$\lambda = 2p - 1.$$

Then

$$P^t = S \begin{pmatrix} 1 & 0 \\ 0 & \lambda^t \end{pmatrix} S^{-1}.$$

Substituting S and S^{-1} , we obtain

$$P^t = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \lambda^t \end{pmatrix} \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

First multiply the first two matrices:

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \lambda^t \end{pmatrix} = \begin{pmatrix} 1 & \lambda^t \\ 1 & -\lambda^t \end{pmatrix}.$$

Therefore,

$$P^t = \frac{1}{2} \begin{pmatrix} 1 & \lambda^t \\ 1 & -\lambda^t \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Multiplying these two matrices gives

$$P^t = \frac{1}{2} \begin{pmatrix} 1 + \lambda^t & 1 - \lambda^t \\ 1 - \lambda^t & 1 + \lambda^t \end{pmatrix}.$$

Finally, because state 0 corresponds to the first row and first column, the $(0, 0)$ entry is the upper-left entry:

$$P_{0,0}^t = \frac{1 + \lambda^t}{2} = \frac{1 + (2p - 1)^t}{2}.$$

Thus again

$$\boxed{P_{0,0}^t = \frac{1 + (2p - 1)^t}{2}}.$$

Interpretation. The eigenvalue 1 corresponds to the stationary distribution. Since P is symmetric and both rows sum to 1, the stationary distribution is

$$\pi = (1/2, 1/2).$$

The second eigenvalue $2p - 1$ measures how quickly the chain forgets its initial state. When $0 < p < 1$, we have $|2p - 1| < 1$, so

$$(2p - 1)^t \rightarrow 0$$

and therefore

$$P_{0,0}^t \rightarrow \frac{1}{2}.$$

Problem 2

Problem. In the one-dimensional random-walk algorithm for the 2-SAT problem, consider a modified random walk with a partially reflecting boundary at 0. Whenever position 0 is reached, with probability 1/2 the walk moves to position 1, and with probability 1/2 the walk stays at 0. Everywhere else, the random walk moves either up or down by 1, each with probability 1/2. Find the expected number of moves to reach n , starting from position 0.

Solution

Let h_i denote the expected number of moves needed to reach n , starting from position i . We want to compute h_0 .

The target state is n , so

$$h_n = 0.$$

For $1 \leq i \leq n-1$, the walk moves to $i-1$ with probability 1/2 and to $i+1$ with probability 1/2. Therefore,

$$h_i = 1 + \frac{1}{2}h_{i-1} + \frac{1}{2}h_{i+1}, \quad 1 \leq i \leq n-1.$$

Equivalently,

$$h_{i+1} - 2h_i + h_{i-1} = -2.$$

This is a second-order difference equation. A quadratic function is natural here. We try

$$h_i = -i^2 + ai + b.$$

Indeed,

$$h_{i+1} - 2h_i + h_{i-1} = -2,$$

so this is the general form of the solution.

Now use the boundary conditions. First,

$$h_n = 0.$$

Thus

$$-n^2 + an + b = 0.$$

Second, at position 0, one move is made. With probability 1/2 the walk stays at 0, and with probability 1/2 it moves to 1. Hence

$$h_0 = 1 + \frac{1}{2}h_0 + \frac{1}{2}h_1.$$

Multiplying by 2 gives

$$2h_0 = 2 + h_0 + h_1,$$

so

$$h_0 = 2 + h_1.$$

Using $h_i = -i^2 + ai + b$, we have

$$h_0 = b$$

and

$$h_1 = -1 + a + b.$$

Therefore,

$$b = 2 + (-1 + a + b) = 1 + a + b.$$

So

$$a = -1.$$

Now apply $h_n = 0$:

$$0 = -n^2 - n + b.$$

Thus

$$b = n^2 + n.$$

Therefore

$$h_i = -i^2 - i + n^2 + n.$$

In particular,

$$\boxed{h_0 = n^2 + n.}$$

Thus, starting from position 0, the expected number of moves needed to reach n is

$$\boxed{n^2 + n.}$$

Comparison with the completely reflecting walk. For the usual completely reflecting boundary, position 0 always moves to 1 in one step, giving the boundary equation $h_0 = 1 + h_1$. That version yields $h_0 = n^2$. The partially reflecting boundary adds an expected delay at 0, giving $h_0 = n^2 + n$.

Problem 3(a)

Problem. Prove that, for every integer n , there exists a coloring of the edges of the complete graph K_n by two colors so that the total number of monochromatic copies of K_4 is at most

$$\binom{n}{4} 2^{-5}.$$

Solution

We use the probabilistic method.

Color every edge of K_n independently red or blue, each with probability $1/2$. For every subset $S \subseteq V(K_n)$ with $|S| = 4$, let X_S be the indicator random variable for the event that the copy of K_4 induced by S is monochromatic. That is,

$$X_S = \begin{cases} 1, & \text{if all edges in } K_n[S] \text{ have the same color,} \\ 0, & \text{otherwise.} \end{cases}$$

A copy of K_4 has

$$\binom{4}{2} = 6$$

edges. It is monochromatic if all six edges are red or all six edges are blue. Hence

$$\Pr[X_S = 1] = \Pr[\text{all six edges are red}] + \Pr[\text{all six edges are blue}] = \left(\frac{1}{2}\right)^6 + \left(\frac{1}{2}\right)^6 = 2^{-5}.$$

Equivalently,

$$\mathbb{E}[X_S] = 2^{-5}.$$

Let

$$X = \sum_{S \in \binom{V(K_n)}{4}} X_S$$

be the total number of monochromatic copies of K_4 . By linearity of expectation,

$$\mathbb{E}[X] = \sum_{S \in \binom{V(K_n)}{4}} \mathbb{E}[X_S] = \binom{n}{4} 2^{-5}.$$

Since $\mathbb{E}[X]$ is the average number of monochromatic K_4 's over all two-color edge-colorings of K_n , there must exist at least one coloring with

$$X \leq \mathbb{E}[X].$$

Therefore there exists a coloring of the edges of K_n by two colors such that the total number of monochromatic copies of K_4 is at most

$$\boxed{\binom{n}{4} 2^{-5}.$$

Problem 3(b)

Problem. Give a randomized algorithm for finding a coloring with at most

$$\binom{n}{4} 2^{-5}$$

monochromatic copies of K_4 that runs in expected time polynomial in n .

Solution

The proof from part (a) suggests the following rejection-sampling algorithm.

Algorithm

1. Randomly color every edge of K_n red or blue independently with probability $1/2$ each.
2. Count the number X of monochromatic copies of K_4 in the resulting coloring.
3. If

$$X \leq \binom{n}{4} 2^{-5},$$

output this coloring.

4. Otherwise, repeat from Step 1.

It remains to show that the expected running time is polynomial in n .

Let

$$N = \binom{n}{4}$$

be the number of copies of K_4 in K_n , and let

$$\mu = \mathbb{E}[X] = N 2^{-5} = \frac{N}{32}.$$

Let

$$p_s = \Pr[X \leq \mu]$$

be the success probability of one trial. We need a lower bound on p_s .

Since X is an integer-valued random variable, the failure event $X > \mu$ implies

$$X \geq \lfloor \mu \rfloor + 1.$$

Thus

$$\mu = \mathbb{E}[X] \geq (\lfloor \mu \rfloor + 1) \Pr[X > \mu].$$

Therefore

$$\Pr[X > \mu] \leq \frac{\mu}{\lfloor \mu \rfloor + 1}.$$

Hence

$$p_s = 1 - \Pr[X > \mu] \geq 1 - \frac{\mu}{\lfloor \mu \rfloor + 1} = \frac{\lfloor \mu \rfloor + 1 - \mu}{\lfloor \mu \rfloor + 1}.$$

Now $\mu = N/32$. Hence the quantity $\lfloor \mu \rfloor + 1 - \mu$ is at least $1/32$: if μ is an integer, then it equals 1; otherwise its fractional gap has denominator 32 and is at least $1/32$. Therefore

$$p_s \geq \frac{1}{32(\lfloor \mu \rfloor + 1)} \geq \frac{1}{32(\mu + 1)} = \frac{1}{N + 32}.$$

Consequently, the expected number of trials before success is at most

$$\frac{1}{p_s} \leq N + 32 = O(n^4).$$

Now consider the running time of one trial. We can count the monochromatic copies of K_4 by enumerating all $\binom{n}{4} = O(n^4)$ subsets of four vertices. For each subset, we inspect its six edges and check whether all six edges have the same color. This takes constant time per subset. Therefore one trial takes

$$O(n^4)$$

time.

The expected total running time is therefore

$$O(n^4) \cdot O(n^4) = O(n^8),$$

which is polynomial in n .

Thus the randomized algorithm finds a desired coloring in expected polynomial time.

Remark. For $n < 4$, there are no copies of K_4 , so every coloring trivially satisfies the required condition. The analysis above is relevant for $n \geq 4$.